



Block error rate performance analysis of RS coded M-QAM modulated coherent OFDM-FSO system

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Abstract

This paper investigates the performance analysis of RS-coded and un-coded coherent M-QAM modulated OFDM free space optical communication system. The system performance is analyzed using block error rate. Mathematical expressions are derived for the average block error rate under the combined effects of attenuation loss, pointing error, and gamma-gamma distributed atmospheric turbulence, and results are validated using Monte Carlo simulation. M-QAM modulated OFDM system with RS codes enhances the system performance more than an un-coded system. Coherent OFDM-FSO improves the system performance by overcoming the effects of atmospheric factors.

Keywords Block error rate · Coded system · Turbulence · Coherent · QAM

1 Introduction

Free Space Optical (FSO) Communication, due to its advantages like high speed, large bandwidth, and ease of deployment unlike fiber optical cables, is generously used nowadays to meet the growing data usage demand and overcome the last mile bottleneck. Apart from the advantages, FSO needs a line-of-sight path, which leads to many hindrances in data transmission. The major factors affecting the optical signal quality are atmospheric turbulence, atmospheric attenuation, and misalignment error. Atmospheric turbulence occurs due to the rapid fluctuation in the temperature and pressure which in turn sets the variation in the refractive index of the optical signal. This causes the intensity fluctuation in the optical signal, which is called as Scintillation effect. Absorption and scattering of the beam result in attenuation in the optical beam. Misalignment error or pointing error occurs due to vibration, which makes alignment change between the transceiver. Pointing errors leads to link failure and increases the error rate in the FSO communication system (Anbarasi et al. 2017).

To combat these hindrances many mitigation techniques like hybrid RF/FSO, diversity, relaying, modulation, and coding techniques have been adopted (Bhatnagar et al.

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2016; Miglani et al. 2019). Appropriate modulation and coding schemes improve the performance of the system. ON–OFF shift keying (OOK) is widely adopted in FSO communication systems due to the advantages like bandwidth efficiency, cost-effectiveness, and receiver design is simple (Ho 2005). But the requirement of a fixed threshold to overcome the atmospheric turbulence effect, and poor spectral and power efficiency make OOK inefficient (Ma et al. 2010). Pulse-Position Modulation (PPM) is power efficient and unlike OOK does not require adaptive thresholds (Androutsos et al. 2021). But it is bandwidth inefficient and tight symbol synchronization is required. Subcarrier Intensity Modulation is adopted to overcome inter symbol interference (ISI). In SIM, many modulation techniques can be adopted to reduce the error rate. BPSK-SIM modulation scheme is considered in Odeyemi et al. (2017), and it is suitable for weak turbulence conditions. The orthogonal Frequency Division Multiplexing (OFDM) technique overcomes the fading effect produced by the particles in the atmosphere. OFDM also reduces the burst errors initiated by atmospheric turbulence and it is bandwidth efficient (Wang et al. 2016). In Sangeetha et al. (2022), the modulation techniques like OOK, DPSK, and DQPSK are compared with the M-ary QAM OFDM system, in which OFDM-based communication system's performance is more suitable for long-distance communication than other modulation systems. Optical OFDM (OOFDM) involves two types of detection techniques: direct and coherent. Though Coherent OOFDM (CO-OFDM) has high complexity in transceiver design, it has eventual performance in spectral efficiency and receiver sensitivity. The performance of CO-OFDM is better than the direct detection OFDM-FSO system by 1 dB attenuation of the FSO link for an acceptable BER (Sharma 2014).

When combined with a suitable error correction coding technique, a coherent OFDM system combats the turbulence-induced burst errors. Error-correcting codes add redundant bits to the data to be transmitted to reduce the errors on the receiver side. Reed–Solomon (RS) codes are fast and simple error-correcting codes and it corrects burst errors (Yu et al. 2004). Under strong turbulence conditions, RS codes attain more than 13 dB gain at BER of 10^{-5} ; at weak turbulence, the RS-coded system behaves like a fade-free channel (Zhao et al. 2010). In Al-Barrak et al. (2017), the error correction codes are enhanced using multipath propagation with a CVMC combiner. It shows that for the Rayleigh fading channel as RS codes can correct burst errors, the performance of RS codes is improved than BCH codes. Coherent OFDM-FSO link with Gamma-Gamma distributed channel performance is analyzed with and without the RS coding technique. RS-coded system outperforms the un-coded system. The performance of the BPSK-coded OFDM system is better than QPSK and OOK-modulated systems (Miglani et al. 2019). To Patel et al. (2021), the system's performance improves by introducing the RS codes, which also makes the system power efficient under various weather and turbulence conditions. In Kumar et al. (2021), the ABER performance of BPSK modulated OFDM RoFSO system is analyzed with BCH and RS codes. The results show that RS and uncoded systems are outperformed by BCH-coded systems.

Under the effect of atmospheric turbulence, as FSO transmits blocks of data at the rate of Gbps, the FSO link behaves like a slowly varying channel, which leads to the exploitation of blocks of data bits (Eaves et al. 1977; Zhang et al. 2014). The effective metric to evaluate the data block corruption and for adopting the appropriate coding technique is the average block error rate (ABLER) (Cheng et al. 2014). In Zhang et al. (2014), Cheng et al. (2014), Ninos et al. (2017), various channel models' block error rate performance is studied. Considering the non-zero boresight pointing error in the Gamma-Gamma distribution, the system performance is analyzed Ninos et al. (2018) with OOK modulation. In

Androutsos et al. (2021), BLER performance is carried out for receiver diversity systems with PPM and PAM modulation techniques.

For our work, we considered Gamma-Gamma distributed channel as it fits well for weak to strong turbulence conditions (Tang et al. 2014). A coherent OFDM system with M-QAM modulation is assumed as QAM modulated OFDM-FSO system suits better for high data rate and long-distance communication, and it reduces the error rate in both strong and weak turbulence conditions Singh et al. (2015) when compared with BPSK, QPSK, and M-QAM with OFDM. M-QAM modulated OFDM system is chosen as it increases the data rate and distance in the FSO communication system and it is used in many applications like 5G, smart city applications, and defense applications (Anandkumar et al. 2022). In the literature, nowhere they considered the OFDM system and channel coding technique for the metric BLER. RS codes are applied to the system as the error correction method improves the FSO link performance. The overall performance of the system is analyzed under various atmospheric turbulence conditions, and pointing errors using the metric BLER.

In Sect. 2, the system and channel model are discussed, in Sect. 3, the BLER equation of coherent OFDM FSO system with M-QAM modulation method is derived and then the coding technique is applied. In Sect. 4, numerical analysis is discussed and Sect. 5, concludes the paper.

2 System and channel model

2.1 System model

The block diagram of the system model is shown in Fig. 1. The data in the block diagram represents the blocks of data. Each block has N-bit streams. The N-bit block-sized data is encoded using a GF (2^N) encoder and the encoded data of length (K) is mapped into

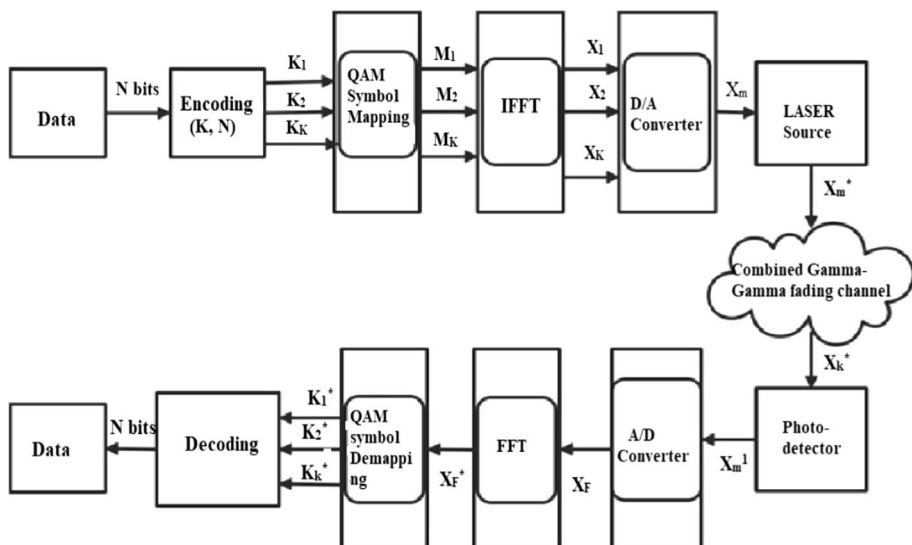


Fig. 1 Block diagram of OFDM-FSO system in terms of Block Error Rate

symbols using a QAM modulator of modulation order $M=2^N$. Based on the block size of data, the selection of encoder code-word length and the modulation order varies. The received signal from the transmitter to the receiver can be written as

$$y = \epsilon hx + n \quad (1)$$

where y is the received signal, x is the transmitted signal, ϵ is the receiver sensitivity, h is the channel gain, and the White Gaussian Noise.

2.2 Channel model

The Gamma-Gamma turbulence channel is considered the FSO turbulence link in the proposed system. The combined channel state which includes, turbulence, atmospheric attenuation and pointing error is given as (Wang et al. 2016)

$$h_{jso} = h_{la} h_{pi} h_t \quad (2)$$

2.2.1 Atmospheric turbulence

To address the weak, moderate, and strong atmospheric turbulence conditions, the Gamma-Gamma channel is used. The probability density function (PDF) of the Gamma-Gamma distribution is given as (Wang et al. 2016)

$$f_{h_t}(h_t) = \frac{2(\alpha\beta)^{(\alpha+\beta)/2}}{\Gamma(\alpha)\Gamma(\beta)} h_t^{(\alpha+\beta-2)/2} K_{\alpha-\beta}\left(2\sqrt{\alpha\beta}h_t\right), h_t > 0 \quad (3)$$

where, $\Gamma(\cdot)$ is the Gamma function, h_t is the scintillation index of an optical signal, and α and β are the small-scale eddies and large-scale intensities of optical wave intensity scintillation. $K_n(\cdot)$ is the modified Bessel function of the second kind of order n . α and β equations are given as

$$\alpha = \exp\left[\left(\frac{0.49\sigma_i^2}{\left(1 + 1.11\sigma_i^{12/5}\right)^{7/6}}\right) - 1\right]^{-1}$$

$$\beta = \exp\left[\left(\frac{0.51\sigma_i^2}{\left(1 + 0.69\sigma_i^{12/5}\right)^{5/6}}\right) - 1\right]^{-1}$$

where σ_i^2 is the Rytov variance given as $\sigma_i^2 = 1.23C_n^2 k^{7/6} L^{11/6}$, wavenumber is $k = 2\pi/\lambda$, link length L and C_n^2 is the refractive index structure coefficient.

2.2.2 Atmospheric attenuation

Attenuation is caused to the propagating light beam in the atmosphere due to the absorption and scattering of light particles. This leads to atmospheric losses. Fog, rain,

and snow are the factors that cause the absorption and scattering of the light beam. The atmospheric loss is given as (Wang et al. 2016)

$$h_{la}(z1) = e^{-(\sigma z1)} \quad (4)$$

where $h_{la}(z1)$ is the attenuation loss encountered by the propagating signal over a distance of $z1$ Kms and σ is the attenuation coefficient.

2.2.3 Pointing error

Pointing error or misalignment error caused by the beam wandering effect results in the radial displacement of the light beam and it is Gaussian distributed as (Wang et al. 2016)

$$h_{pi}(\delta) = A_0 e^{\left(-\frac{2\delta^2}{w_{zeq}^2}\right)} \quad (5)$$

where, t equivalent beam width is $w_{zeq}^2 = w_{z1}^2 \sqrt{\pi} \operatorname{erf}(v) / 2v \exp(-v^2)$, $v = \sqrt{\pi} r / \sqrt{2} w_{z1}$, r is the aperture radius of the receiver. w_{z1} is the beam width at link length of $z1$, the normalized beam width is w_{z1}/r , $A_0 = [\operatorname{erf}(v)]^2$ is the power received at $\delta=0$. The pdf of h_{pi} is written as (Wang et al. 2016)

$$f_{h_{pi}}(h_{pi}) = \frac{\gamma_{pi}^2}{A_0} h_{pi}^{\gamma_{pi}^2-1}, 0 \leq h_{pi} \leq A_0 \quad (6)$$

where, $\gamma_{pi} = w_{zeq}^2 / 2\delta_s$, δ_s is the jitter, and δ_s/r is the normalized jitter error.

The combined channel effect is represented as (Tang et al. 2014)

$$f_{h_{fso}}(h_{fso}) = \int f_{h_{fso}h_t}(h_{fso}/h_t) f_{h_t}(h_t) dh_t \quad (7)$$

$$f_{h_{fso}h_t}(h_{fso}/h_t) = \frac{1}{h_{la}h_t} f\left(\frac{h_{fso}}{h_{la}h_t}\right) = \frac{\gamma_{pi}^2}{A_0^{\gamma_{pi}^2} h_{la}h_t} \left(\frac{h_{fso}}{h_{la}h_t}\right)^{\gamma_{pi}^2-1} \quad 0 \leq h_{fso} \leq A_0 h_{la}h_t \quad (8)$$

By substituting Eqs. (3 and 5) into Eq. (7), we get (Wang et al. 2016)

$$f_{h_{fso}}(h_{fso}) = \frac{\alpha \beta \gamma_{pi}^2}{A_0 h_{la} \Gamma(\alpha) \Gamma(\beta)} G_{1,3}^{3,0} \left[\frac{\alpha \beta h_{fso}}{A_0 h_{la}} \middle| \gamma_{pi}^2, \alpha-1, \beta-1 \right] \quad (9)$$

here, $G_{p,q}^{m,n}$ denotes the Meijer-G-function (Wang et al. 2015). The PDF of (9) in terms of SNR can be written as (Tang et al. 2014).

$$f_{\mu_h}(\mu_h) = \frac{\gamma_{pi}^2}{\mu_h \Gamma(\alpha) \Gamma(\beta)} G_{1,3}^{3,0} \left[\frac{\alpha \beta \mu_h}{(1 + \gamma_{pi}^2) \bar{\mu}_h} \middle| \gamma_{pi}^2 + 1, \alpha, \beta \right] \quad (10)$$

where μ_h is the instantaneous electrical SNR at the receiver side, $\bar{\mu}_h$ is the average electrical SNR.

2.3 Reed–Solomon (RS) coding and decoding

Reed–Solomon codes are good for burst error and random error correction as these codes work on the symbol level. Several bits form a symbol. The generator polynomial $g(x)$ of length $n = 2^m - 1$ and t -error correcting codes is (Lin, 2001)

$$g(x) = (x - \alpha)(x - \alpha^2) \dots (x - \alpha^{2^t}) \quad (11)$$

where α is the primitive elements of $GF(2^m)$ and ‘ m ’ is the number of bits per symbol. The encoded codeword $c(x)$ is represented as $c(x) = m(x) \times x^{n-k} + r(x)$. The encoding process involves the multiplication of the message by x^{n-k} and then divided by generator polynomial $g(x)$, to obtain the quotient as $q(x)$ and remainder as $r(x)$. In the decoding process, using the Berlekamp–Massey algorithm the error location polynomial is computed and Forney’s algorithm is used to find the magnitude of errors.

In this paper, RS code word length n is chosen as 16 and 64 depending on the block size of data as $N=4$ and 6 bits. The error-correcting capability t is fixed as 2 for both the code word lengths.

3 Average block error rate (ABLER)

BLER is defined as the probability of occurrence of more than M -bit error detections, which is binomially distributed in a block of N bits, and it is given as (Ninos et al. 2018).

$$P(M, N; \mu_h) = \sum_{m=M+1}^N \binom{N}{m} P_b^m (1 - P_b)^{N-m} \quad (12)$$

where P_b is the bit error probability of the OFDM system with the M-QAM modulation method and is given as (Wang et al. 2015)

$$P_{OFDM} = \frac{4}{\log_2 M_O} Q \left(\sqrt{\frac{3TB \log_2 M_O}{N_O(M_O - 1)}} \mu_h \right) \quad (13)$$

where is the instantaneous SNR. OFDM parameters are the number of subcarriers (N_O), bandwidth (B), and symbol period (T), M_O is the QAM modulation index.

The average BLER of the system can be written as (Cheng et al. 2014)

$$P_{BLER_OFDM} = \int_0^\infty f_{\mu_h}(\mu_h) P(M, N; \mu_h) d\mu_h \quad (14)$$

The binomial expansion of $P_b^m (1 - P_b)^{N-m}$ is represented as (Ninos et al. 2018)

$$\sum_{k_1=0}^{N-m} \binom{N-m}{k_1} (-1)^{k_1} (P_b)^{m+k_1} \quad (15)$$

Substituting the Eqs. (12 and 15) in Eq. (14), we get the equation as

$$\int_0^\infty f_{\mu_h}(\mu_h) \sum_{m=M+1}^N \binom{N}{m} \sum_{k_1=0}^{N-m} \binom{N-m}{k_1} (-1)^{k_1} \left(\frac{4}{\log_2 M_O} Q \left(\sqrt{\frac{3TB \log_2 M_O}{N_O(M_O-1)} \mu_h} \right) \right)^{m+k} d\mu_h \quad (16)$$

From (Ninos et al. 2018), the approximate expression for estimating $Q(x)$ is

$$Q(x) \simeq \left[5 \exp(-2x^2) + 4 \exp\left(-11/20x^2\right) + \exp\left(-1/2x^2\right) \right] / 24 \quad (17)$$

Equation (16) now becomes

$$\begin{aligned} P_{BLER_{OFDM}} = & \int_0^\infty f_{\mu_h}(\mu_h) \sum_{m=M+1}^N \binom{N}{m} \sum_{k_1=0}^{N-m} \binom{N-m}{k_1} (-1)^{k_1} \\ & \left(\frac{4}{\log_2 M_O} \times \left(\frac{5}{24} \exp\left(-\frac{\mu_h^2 6TB \log_2 M_O}{N_O(M_O-1)} \gamma\right) \right. \right. \\ & \left. \left. + \frac{1}{6} \exp\left(-\frac{\mu_h^2 33TB \log_2 M_O}{20N_O(M_O-1)} \gamma\right) + \frac{1}{24} \exp\left(-\frac{\mu_h^2 3TB \log_2 M_O}{2N_O(M_O-1)} \gamma\right) \right) \right)^{m+k} d\mu_h \end{aligned} \quad (18)$$

Considering, $x_1 = \frac{3TB \log_2 M_O}{N_O(M_O-1)}$ and using the multinomial theorem in Eq. (18)

$$\begin{aligned} P_{BLER_{OFDM}} = & \int_0^\infty f_{\mu_h}(\mu_h) \sum_{m=M+1}^N \binom{N}{m} \sum_{k_1=0}^{N-m} \binom{N-m}{k_1} \\ & \sum_{t=0}^{m+k} \binom{m+k}{t} \sum_{l=0}^t \binom{t}{l} (-1)^{k_1} \left(\frac{1}{24} \right)^{m+k-t} \left(\frac{5}{24} \right)^{t-l} \left(\frac{1}{6} \right)^l \\ & \times \exp\left(-x_1 \mu_h \left(\frac{m+k+3t}{2} - \frac{29l}{20} \right)\right) d\mu_h \end{aligned} \quad (19)$$

Substituting Eq. (9) and Meijer-G-function of exponential form in Eqs. (19 and 20)

$$\begin{aligned} P_{BLER_{OFDM}} = & \sum_{m=M+1}^N \sum_{k_1=0}^{N-m} \sum_{t=0}^{m+k} \sum_{l=0}^t \binom{N}{m} \binom{N-m}{k_1} (-1)^{k_1} \\ & \times \left(\frac{1}{24} \right)^{m+k-t} \binom{t}{l} \left(\frac{5}{24} \right)^{t-l} \left(\frac{1}{6} \right)^l \frac{\gamma_{pi}^2}{\mu_h \Gamma(\alpha) \Gamma(\beta)} \\ & \times \int_0^\infty G_{1,3}^{3,0} \left[\frac{\alpha \beta \mu_h}{(1 + \gamma_{pi}^2) \mu_h} \middle| \gamma_{pi}^2 + 1, \alpha, \beta \right] \\ & \times G_{0,1}^{1,0} \left[x_1 \mu_h \left(\frac{m+k+3t}{2} - \frac{29l}{20} \right) \middle| \cdot \right] d\mu_h \end{aligned} \quad (20)$$

Simplifying Eq. (20) by performing the integration of Meijer-G-function (Adamchik et al. 1990), the Average BLER of the coherent OFDM system with M-QAM modulation using the combined effect of Gamma-Gamma distributed channel is obtained as

$$\begin{aligned}
P_{BLER_{OFDM}} &= \sum_{m=M+1}^N \sum_{k_1=0}^{N-m} \sum_{t=0}^{m+k_1} \sum_{l=0}^t \binom{N}{m} \binom{N-m}{k_1} (-1)^{k_1} \\
&\times \binom{m+k_1}{t} \left(\frac{1}{24}\right)^{m+k_1-t} \binom{t}{l} \left(\frac{5}{24}\right)^{t-l} \left(\frac{1}{6}\right)^l \frac{\gamma_{pi}^2 2^{\alpha+\beta-3}}{\pi \Gamma(\alpha) \Gamma(\beta)} \\
&\times G_{3,2}^{1,3} \left[\frac{x_1 \mu_c (\gamma_{pi}^2 + 1)^2}{\alpha \beta (\gamma_{pi}^2)^2} \middle| \begin{matrix} 1 - \gamma_{pi}^2, 1 - \alpha, 1 - \beta \\ 0, -\gamma_{pi}^2 \end{matrix} \right]
\end{aligned} \quad (21)$$

where $\mu_c = \overline{\mu}_h \left(\frac{m+k_1+3t}{2} - \frac{29l}{20} \right)$

3.1 ABLER of RS-coded system

The probability of error after decoding (n, k, t) RS code over GF (2^m) for t symbol error correcting capability is given as (Naik et al. 2020)

$$P_{ABLER_{RS}} \leq \sum_{i=t+1}^n \binom{n}{i} p_s^i (1 - p_s)^{n-i} \quad (22)$$

where $p_s = 1 - (1 - p_b)^m$, p_b is the system error probability. $P_{ABLER_{RS}}$ is obtained by placing Eq. (20) in the place of p_b .

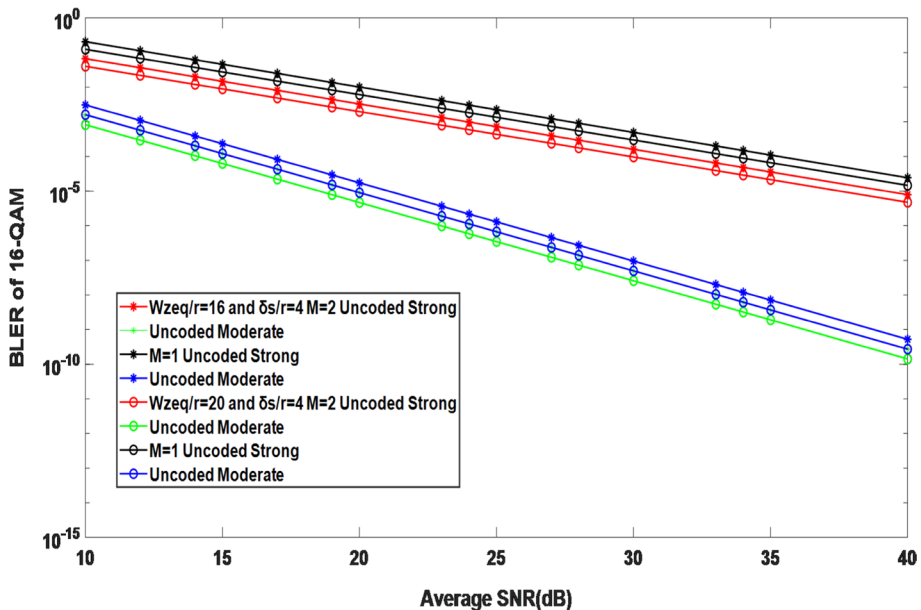


Fig. 2 BLER of uncoded 16-QAM OFDM FSO system under different pointing errors and $M=1$ and 2

4 Numerical results and discussion

In this section, the numerical results are analyzed for the parameter average block error rate under the atmospheric turbulence and pointing error effect of the QAM modulated OFDM system. The analyses are carried out with $\alpha = 4.34$ and $\beta = 1.31$ for strong turbulence conditions and $\alpha = 4.19$ and $\beta = 2.26$ for moderate turbulence conditions (Wang et al. 2016). For the pointing error effect, $w_{zeq}/r = 16$ and 20 , $\delta_s/r = 4$ is considered. The modulation order M_O of the QAM modulation technique is considered as 16 and 64 with $N_O = 256$. The block length N is assumed as 4 and 6 bits and error-detecting probabilities M of more than 2 and 1 are considered. Depending on the block length N , the modulation order and the RS code word length vary = 4 bits, the RS (15, 11) code and $M_O = 16$ is taken. For $N = 6$ bits, the RS (63, 59) code and $M_O = 64$ is taken for analysis. For both the code word lengths, the error correction capability t is considered as 2.

Figure 2. shows the BLER performance of the 16-QAM modulated OFDM system when the pointing error and turbulence conditions vary with the variation in the number of error bits that occur in the system. More faulty blocks are detected when the number of error bits M in a block is more than 1 than the error bits more than $M = 2$. When the pointing error is $w_{zeq}/r = 16$ and $\delta_s/r = 4$ the BLER is more. The system deteriorates more for the high pointing error. Performance of the system is better for low pointing error ($w_{zeq}/r = 20$ and $\delta_s/r = 4$) when compared with high pointing error ($w_{zeq}/r = 16$ and $\delta_s/r = 4$). An increase in the turbulence weakens the system performance and imparts more error to the communication system.

BLER of 64-QAM OFDM system is shown in Fig. 3. The block size of data taken for analysis is 6, so the modulation order of QAM is 64 (2^6). Compared with Fig. 2, the BLER is high when the modulation order increases from 16 to 64. At 20 dB SNR, the BLER of the system with pointing error as $w_{zeq}/r = 16$ and $\delta_s/r = 4$ is 10^{-2} and 10^{-5} for $M = 2$ and

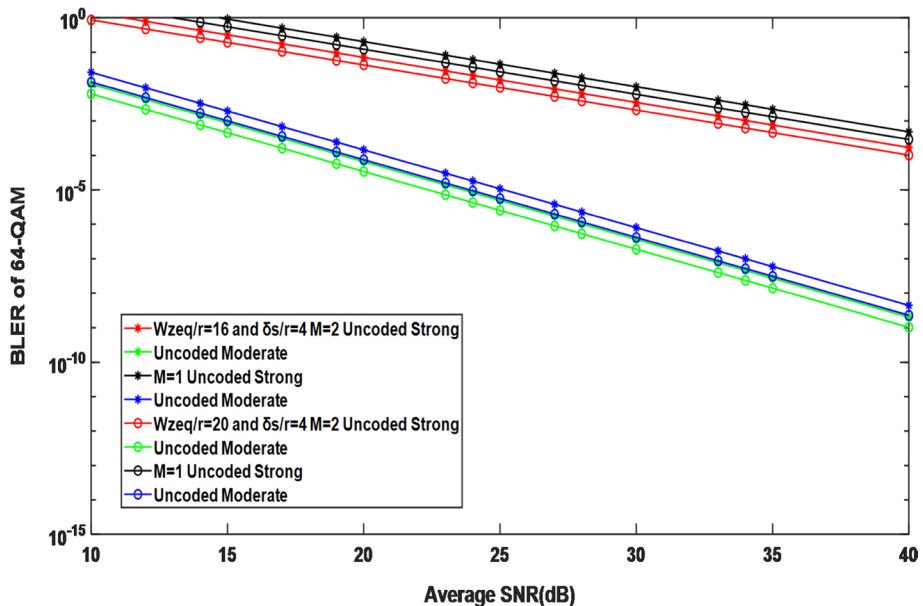


Fig. 3 BLER of un-coded 64-QAM OFDM FSO system under different pointing errors and $M = 1$ and 2

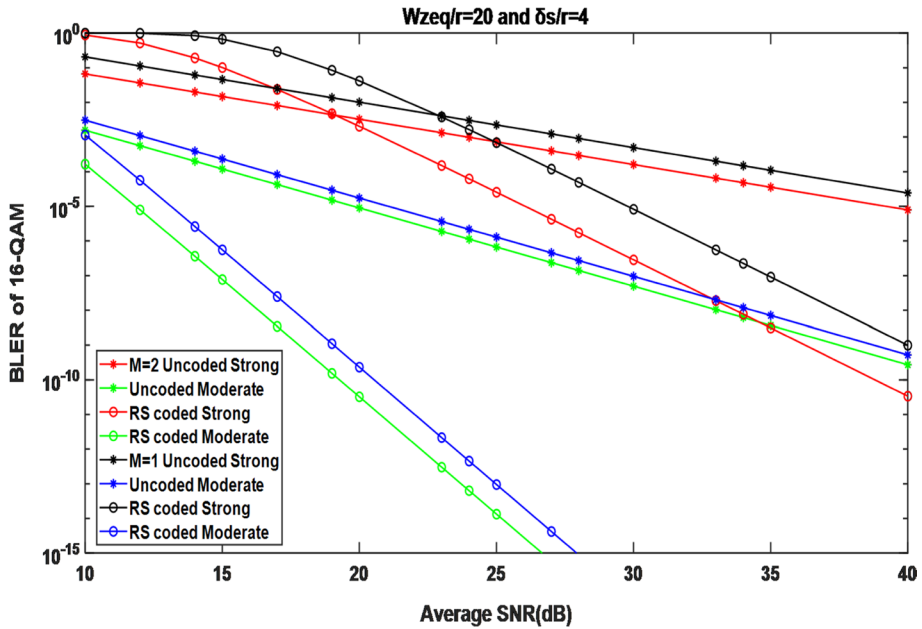


Fig. 4 BLER of un-coded and RS (15, 11, 2) coded 16-QAM OFDM FSO system for $w_{zeq}/r = 20$, $M = 1$ and 2

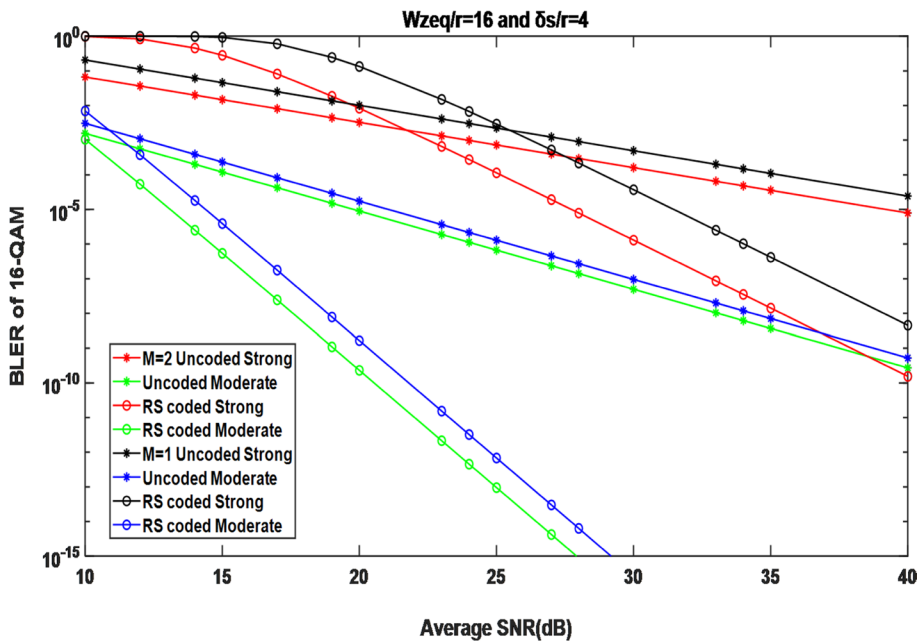


Fig. 5 BLER of un-coded and RS (15, 11, 2) coded 16-QAM OFDM FSO system for $w_{zeq}/r = 16$, $M = 1$ and 2

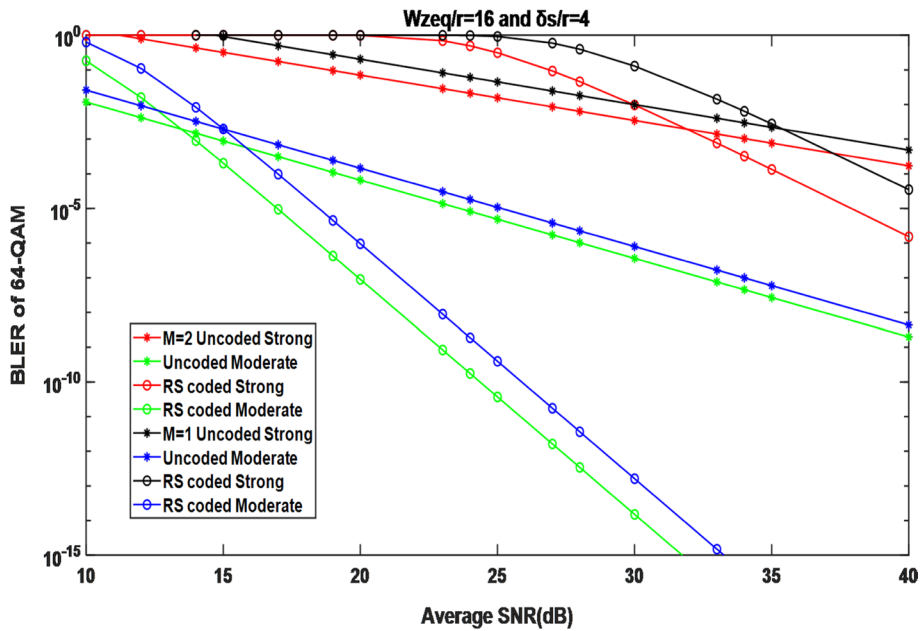


Fig. 6 BLER of un-coded and RS (63, 59, 2) coded 64-QAM OFDM FSO system for $w_{zeq}/r = 16$, $M=1$ and 2

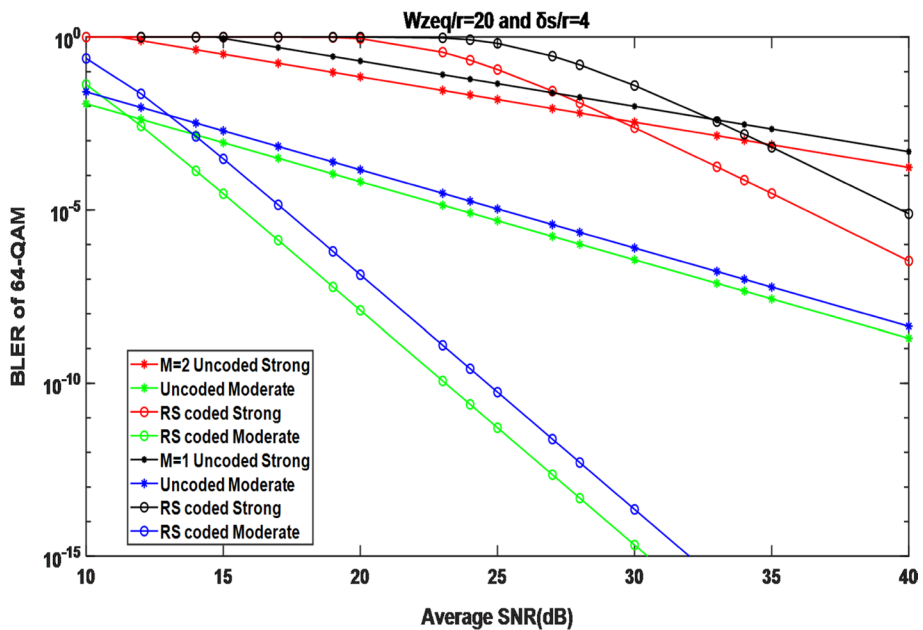


Fig. 7 BLER of un-coded and RS (63, 59, 2) coded 64-QAM OFDM FSO system for $w_{zeq}/r = 20$, $M=1$ and 2

10^{-1} and 10^{-4} for $M=1$ bit error for strong and moderate turbulence conditions respectively. As the probability of finding bit error increases, more errors are detected in a block of data. When the pointing error is reduced by increasing the beam width, the BLER is reduced. The error rate in the un-coded system depends on the pointing error value and the block size of the data. The more the number of bits in a block more the probability of occurrence of errors.

Figures 4 and 5, show the BLER performance comparison of un-coded and RS-coded 16-QAM OFDM-FSO system under different turbulence, pointing error and error bits M . The block size of the data $N=4$. The code word length of the RS code is chosen from the number of bits in the block as $GF(2^N)$. RS code of length (15, 11, 2) is used for $N=4$ bits and the order of modulation for QAM is taken as ($2^{\text{Block size}} = 16$). Here one code word of the RS code is mapped to one symbol of QAM. The influence of the turbulence is more in the system. In both the high and low pointing error values, for strong turbulence conditions beyond 25 dB SNR only, the coded system performs well than the un-coded system. The RS codes reduce the burst error at low SNR values for the moderate turbulence condition.

RS-coded 64-QAM OFDM-FSO system BLER performance is shown in Figs. 6 and 7. The results show that an increase in the order of modulation increases the BLER. The block size for analyzing the 64-QAM OFDM system is assumed as 6 bits. So, the RS code of (63, 59, 2) is chosen as the error correction code. 16-QAM gives lower BLER than 64-QAM. High SNR is required for the RS-coded 64-QAM system to reduce the errors. For strong turbulence condition, RS codes reduce error in the system beyond 35 dB. SNR gain of 10 dB is achieved by the 16-QAM system than the 64-QAM system.

Figures 8 and 9 show the BLER of the un-coded OFDM FSO system for various modulation orders of QAM. BLER of the system increases, with the increase in the length of N

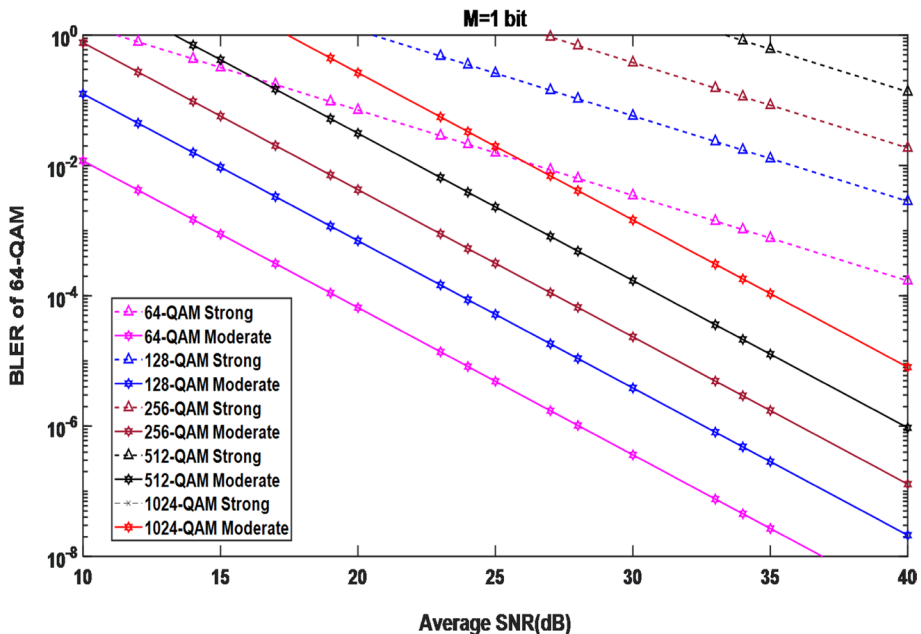


Fig. 8 BLER of un-coded QAM OFDM FSO system under various orders of M_O and $M=1$ bit

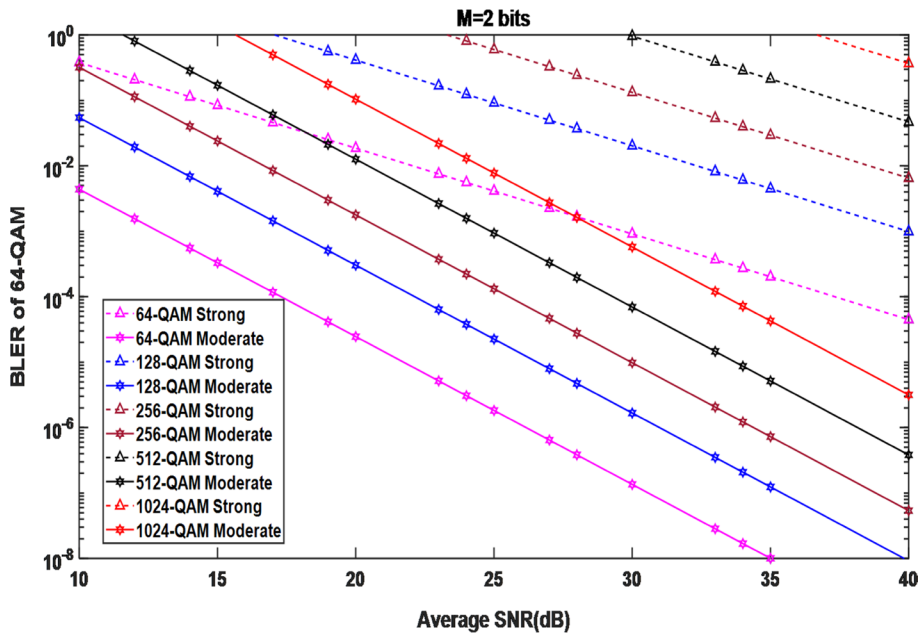


Fig. 9 BLER of un-coded QAM OFDM FSO system under various orders of M and $M=2$ bits

Table 1 BLER of M-QAM modulated OFDM-FSO system under strong turbulence conditions at SNR 40 dB

M-QAM	16-QAM				64-QAM			
	M=2		M=1		M=2		M=1	
	Un-coded	RS-coded	Un-coded	RS-coded	Un-coded	RS-coded	Un-coded	RS-coded
$w_{zeq}/r = 16$ and $\delta_s/r = 4$	7×10^{-6}	1×10^{-10}	2×10^{-5}	4×10^{-9}	1.7×10^{-4}	1×10^{-6}	4×10^{-4}	3×10^{-5}
$w_{zeq}/r = 20$ and $\delta_s/r = 4$	7.8×10^{-6}	3×10^{-11}	2.4×10^{-5}	9×10^{-10}	1×10^{-4}	3×10^{-7}	4.8×10^{-4}	7×10^{-6}

bits, increasing the modulation order of the QAM, which decreases the performance of the system.

Table 1 gives the BLER performance comparison of M-QAM OFDM system in terms of modulation order, the number of error bits M , and pointing error values under strong turbulence conditions at 40 dB SNR. It is shown that RS coded system gives better performance than the un-coded OFDM system at high SNR values for strong turbulence conditions. For moderate turbulence conditions, the RS-coded system enhances the performance of the un-coded system.

5 Conclusion

The performance of a coherent OFDM-FSO system with QAM modulation is investigated in this paper with RS codes. The code word length of the RS codes and the modulation order of M-QAM is not fixed and the values are taken depending on the block size of the transmitted data. 16-QAM with RS (15, 11) is taken and 64-QAM with RS (63, 59) is chosen for error reduction for the block size of data $N=4$ and 6 bits respectively. RS codes reduce burst errors in the system. The performance of the system deteriorates under the strong turbulence condition and number of the erroneous bits in blocks of data.

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Declarations

Conflict of interest Authors have no competing interests.

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